Zomato Restaurant Data Analysis Report

1. Introduction

The **Zomato Restaurant Data Analysis** project aims to explore restaurant data to understand customer preferences, restaurant performance, pricing trends, and location-based insights. Using **Exploratory Data Analysis (EDA)**, the project identifies patterns that can help restaurant owners and businesses make informed decisions to improve customer satisfaction and business growth.

2. Dataset Overview

The dataset was collected from **Zomato** and contains restaurant information such as names, locations, cuisines, ratings, votes, pricing, online ordering, and table booking facilities.

Dataset Summary

- Original Records: **34,818**
- Cleaned Records: **23,918**
- Total Features: **13**

3. Data Preprocessing

The dataset was cleaned to improve data quality before analysis.

The preprocessing steps included:

- Removing duplicate records.
- Handling missing values.
- Converting ratings, votes, and cost into numerical format.
- Creating **Main Cuisine** and **Price Category** features.
- Categorizing restaurants into Budget, Moderate, Expensive, and Luxury groups.

4. Exploratory Data Analysis

The following analyses were performed:

- Restaurant Rating Distribution
- Popular Cuisine Analysis
- Cost Distribution
- Restaurant Location Analysis

- Correlation Heatmap
- Online Order vs Rating
- Table Booking vs Rating
- Cuisine WordCloud

These visualizations helped identify customer preferences and restaurant trends.

5. Key Findings

The analysis revealed several important insights:

- Most restaurants have ratings between **3.5 and 4.2**.
- **North Indian**, **South Indian**, and **Cafe** are the most popular cuisines.
- Budget and Moderate-priced restaurants dominate the market.
- **BTM**, **HSR Layout**, and **JP Nagar** have the highest number of restaurants.
- Restaurants with online ordering and table booking generally receive higher customer ratings.
- Higher-rated restaurants tend to receive more customer votes.

6. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Expand businesses in high-demand locations.
- Maintain affordable pricing to attract more customers.
- Promote online ordering and table booking services.
- Focus on customer feedback to improve ratings.
- Use both popular and niche cuisines to reach a wider audience.

7. Conclusion

This project demonstrates how **Exploratory Data Analysis (EDA)** can provide valuable insights into restaurant performance and customer behavior. The findings show that factors such as location, cuisine, pricing, and online services significantly influence restaurant success. These insights can help restaurant owners and food businesses make data-driven decisions to improve customer satisfaction and business growth.